

CST2213: Business Intelligence Programming 2

Ottawa Neighbourhood Safety Clusters

Project Phase 2 – Data Preparation and Initial Analysis

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Project Overview

This report documents *Phase 2* of the project: data preparation and initial analysis. The goal of the project is to segment Ottawa's neighbourhoods into distinct safety profiles by combining several municipal open-data crime datasets with police calls-for-service data, then applying KMeans unsupervised clustering. Phase 2 covers everything from raw data ingestion through cleaning, integration, exploratory analysis, feature preparation, and the first modelling results that validate the approach.

1. Data Cleaning and Preprocessing

Ensured data quality across all seven datasets by handling missing values, duplicates, inconsistent naming, and irrelevant records, producing a single analysis-ready table aggregated by neighbourhood.

1.1 Cleaning steps taken

Per-dataset aggregation

Each raw dataset was aggregated to a count of incidents per neighbourhood (the unit of analysis). Specific handling by dataset:

- Auto Theft - counted per neighbourhood; records flagged "Out of Jurisdiction" were dropped.
- Bike Theft - counted per neighbourhood.
- Criminal Offences - counted per neighbourhood × offence category and pivoted wide (18 offence types). The "Theft – Motor Vehicle" category was dropped because it duplicates the dedicated Auto Theft dataset. "Carp" and "Carp Ridge" were consolidated into a single row (same neighbourhood, inconsistent labels).
- Hate Crime - counted per neighbourhood × motivation and pivoted wide. Four ambiguous/indeterminate categories (Combination, Other Similar Factor, Unknown, Not Applicable) were collapsed into a single "Other Hate Crimes" column.
- Homicide & Shootings - counted per neighbourhood.
- Calls for Service - filtered to priority levels 1–3 only (dispatched responses; priorities 4–7 are non-deployed). Priorities were mapped to severity labels: Police_Critical (life-threatening), Police_High (serious harm possible), Police_Medium (risk with delay).
- Neighbourhood boundaries - reduced to ONS_ID (neighbourhood ID), Name, population estimate (POPEST), and geometry.

Handling missing values

After each dataset was left-joined onto the crime base table, neighbourhoods with no recorded incidents of a given type produced nulls. These were filled with 0, since an absent crime record legitimately means zero incidents rather than unknown data.

Handling duplicates and inconsistencies

Duplicate/overlapping records were removed (e.g the Theft – Motor Vehicle example above). Inconsistent neighbourhood naming between the crime datasets and the boundary/calls datasets was the single largest data-quality issue and is documented under Integration.

Categorical-to-numerical conversion

The categorical offence, hate-motivation, and homicide-category fields were converted to numeric feature columns through pivot-and-count encoding: each category became its own column holding the incident count per neighbourhood. This yields an entirely numeric feature matrix suitable for distance-based clustering.

2. Data Integration and Transformation

Merged the seven aggregated datasets into one neighbourhood-level table and transformed it into a structured, model-ready format.

2.1 Integration strategy

The aggregated Criminal Offences table (113 neighbourhoods — the widest coverage) was used as the left base. The remaining crime tables were left-joined onto it on the neighbourhood key, with missing values filled with 0

2.2 Neighbourhood name reconciliation

The crime datasets and the boundary/calls datasets use different neighbourhood naming conventions, only 69 of 113 names matched exactly. Mismatches were resolved in three passes:

- 1. Polygon unions** - crime-dataset names that correspond to several map polygons had their geometries unioned first (e.g. Barrhaven ← Old Barrhaven West + Old Barrhaven East).
- 2. Manual overrides** - a lookup table of confirmed low-confidence matches was applied (e.g. Bayshore → Bayshore – Belltown; CFB Rockcliffe-NRC → Wateridge Village).
- 3. Fuzzy matching** - remaining names were matched with rapidfuzz at a similarity threshold of 70.

The reconciled tables were then inner-joined so that population, geometry, and police-call counts attach to each crime row.

2.3 Feature transformations

- **Feature consolidation** - twelve low-frequency / petty categories (e.g. Fraud, Prostitution, and the hate-motivation columns) were summed into a single "Other" feature to reduce sparsity.
- **Priority encoding** - police-call priorities collapsed into three severity features (Critical / High / Medium).

- **Scope trimming** - nine Greenbelt sub-areas were dropped (ambiguous boundaries, negligible residential population). Remaining unmatched rural/edge names were also removed.

Final integrated dataset: 104 neighbourhoods retained after reconciliation and scope trimming.

3. Exploratory Data Analysis (EDA)

Understand the distribution and relationships in the cleaned data before modelling, using summary statistics and visualisation.

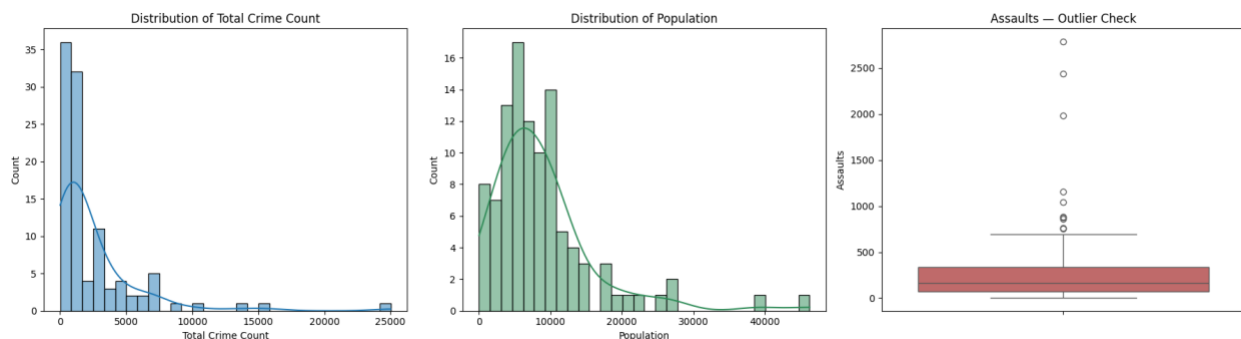
3.1 Summary statistics

	TotalCrime	ViolentCrime	PropertyCrime	Population	Police_Critical	Police_High	Police_Medium
count	104.0	104.0	104.0	104.0	104.0	104.0	104.0
mean	2439.2	569.7	1869.5	8736.3	130.5	1686.2	3094.8
std	3540.1	767.7	2809.3	7409.9	212.8	2364.3	4310.5
min	10.0	1.0	6.0	17.0	0.0	9.0	19.0
25%	545.2	152.2	394.5	4245.0	42.5	563.8	949.2
50%	1224.5	326.5	823.5	6805.5	61.5	894.0	1517.5
75%	2895.2	669.8	2336.0	10701.0	138.2	1990.0	3992.5
max	25014.0	5474.0	19540.0	46263.0	1317.0	18859.0	28926.0

Key statistics narrative:

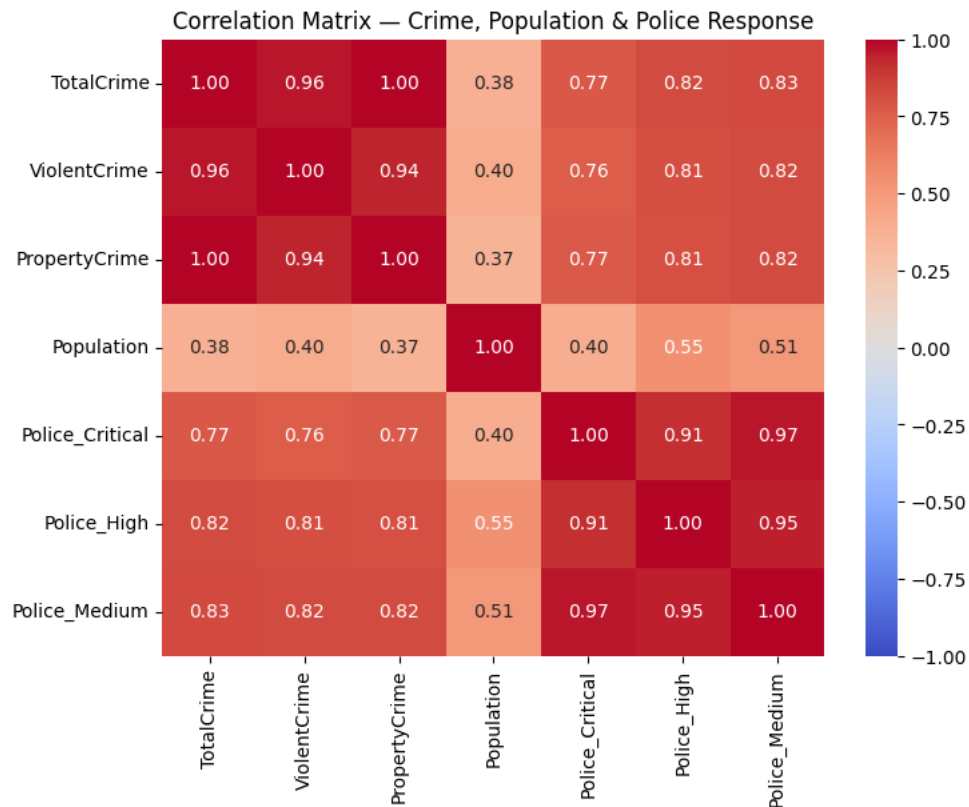
- **The spread is enormous:** TotalCrime ranges from 10 incidents (Notre-Dame Cemetery) to 25,014 (Centretown) — roughly a 2,500x range across 104 neighbourhoods.
- The five highest-crime neighbourhoods (Centretown, Byward Market, Lowertown, Overbrook-McArthur, New Barrhaven-Stonebridge) alone account for a disproportionate share of citywide incidents, consistent with downtown/commercial-core neighbourhoods driving the tail.
- The three police-response features follow the same right-skewed shape as the crime counts, which is expected since response volume tracks incident volume.

3.2 Distributions



Both `TotalCrime` and `Population` show the same long right tail, with most neighbourhoods clustered at low values and a handful of high-crime, high-population outliers stretching the axis. The `Assaults` box plot makes the outlier structure explicit, the bulk of neighbourhoods sit in a tight low-value box, while a small number of points (the same downtown/core neighbourhoods) sit far above the upper whisker.

3.3 Relationships between variables



- `ViolentCrime` and `PropertyCrime` are strongly correlated with each other and with `TotalCrime` - neighbourhoods with more violent crime also tend to have more property crime, i.e. the two crime types largely move together rather than trading off.
- The three police-response features are strongly correlated with each other and moderately-to-strongly correlated with crime counts, confirming response volume tracks incident volume as expected.
- `Population` correlates only weakly-to-moderately with crime, bigger neighbourhoods do not proportionally have more crime, so raw counts conflate neighbourhood size with actual risk. This motivates normalising crime figures per-capita in the analysis/modelling phase rather than relying on raw counts alone.

4. Feature Selection and Engineering

Identify and construct the feature set used for clustering, and justify the choices.

4.1 Engineered features

- **"Other" composite** - twelve sparse crime/hate categories summed into one column to reduce noise and dimensionality.
- **Police severity tiers** - raw priority levels engineered into Police_Critical / High / Medium counts.
- **Per-capita measures (analysis stage)** — crime-per-capita and critical-calls-per-capita derived using neighbourhood population to compare fairly across differently-sized areas.

4.2 Final feature set

The clustering model uses all numeric columns except the Neighbourhood name and geometry, 22 features covering the crime counts, the "Other" composite, population, and the three police-response tiers. Features were standardised with StandardScaler prior to clustering so that no single high-volume feature dominates the Euclidean distance.

4.3 Justification

Each retained feature represents a distinct dimension of neighbourhood safety (violent crime, property crime, hate-motivated incidents, and police-response demand). Standardisation rather than removal was chosen because extreme values in this domain are genuine signal, the busiest neighbourhoods are meaningfully different, not wrong.

5. Preliminary Insights

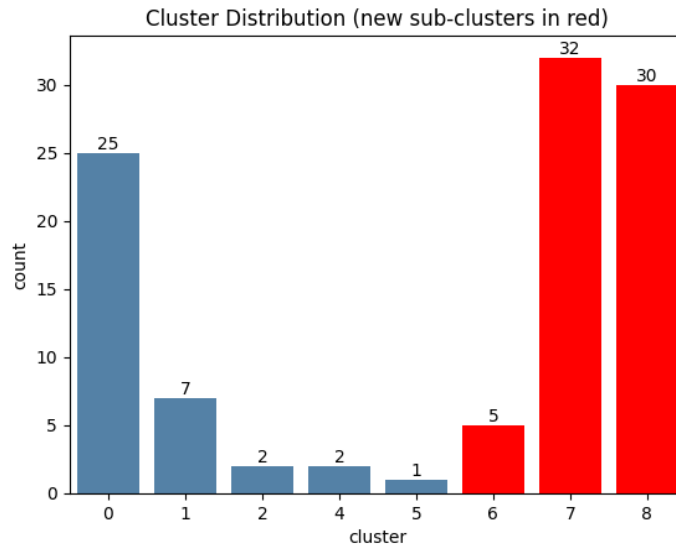
Validated the initial approach with a first modelling pass and early analysis and set direction for the next phase.

5.1 Clustering approach

K-Means clustering was applied to the 22-feature standardised matrix. The number of clusters K was selected using two complementary methods:

- **Elbow method (WCSS)** - within-cluster sum of squares computed for $K = 2 \dots 11$; the curve bends at **$K = 6$** .
- **Silhouette method** - average silhouette score over the same range confirmed $K = 6$ as optimal.

Because the initial cluster 3 absorbed a large share of neighbourhoods, it was sub-clustered (silhouette-guided, sub- $K = 3$) into finer groups, yielding 8 final clusters. The model configuration: k-means++ initialisation, max_iter = 100, random_state = 42 for reproducibility.

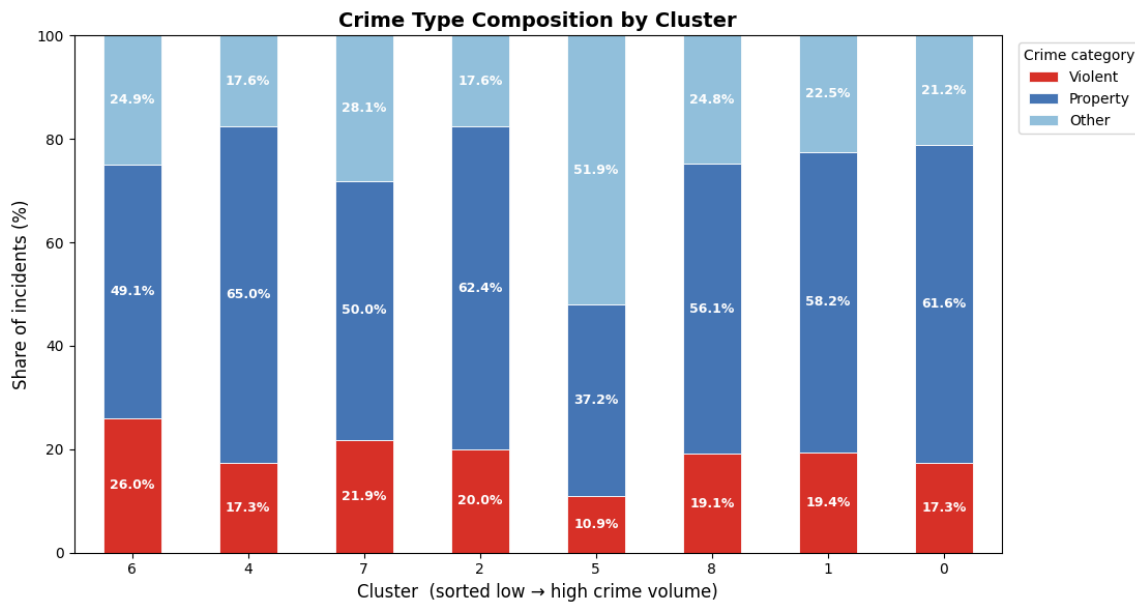


Neighbourhoods per Cluster

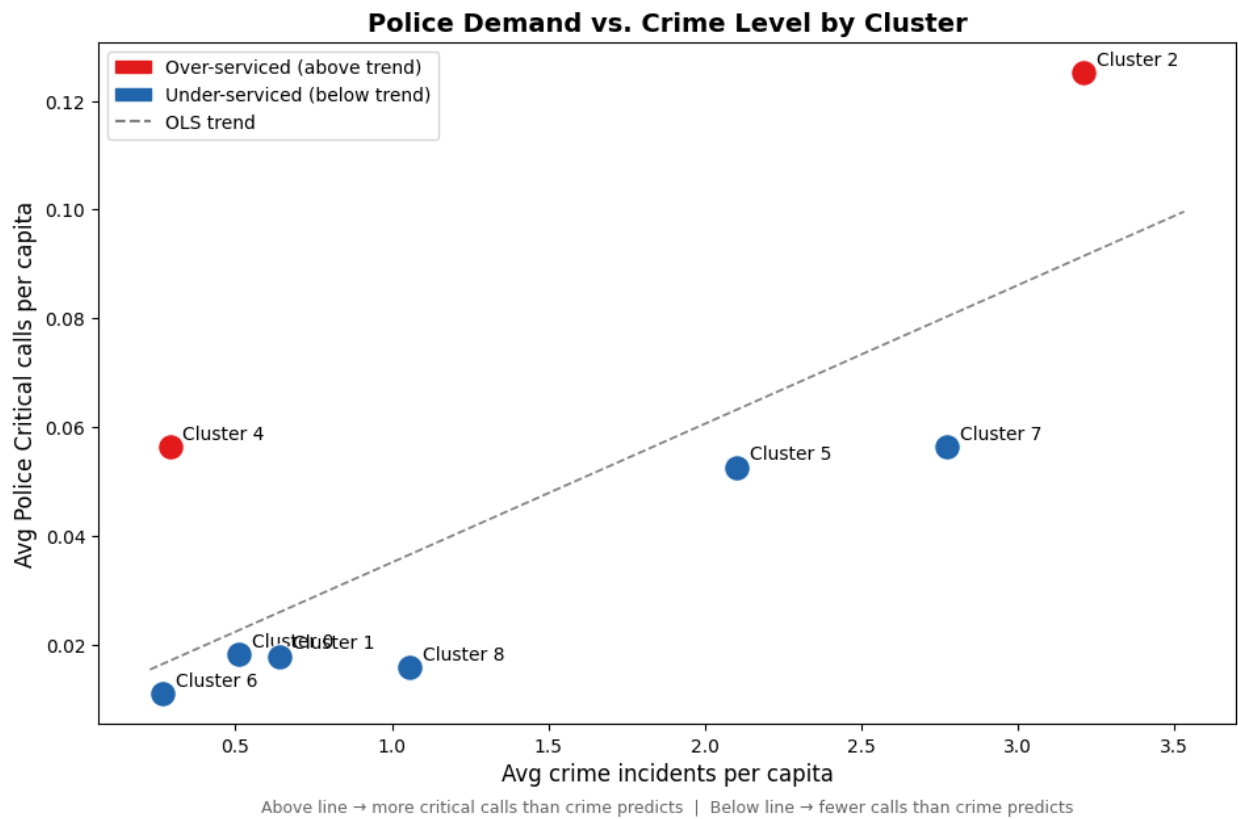
5.3 Preliminary analytical findings

Five analyses were run on the clustered data (full detail reserved for the next phase):

1. **Geospatial map** - neighbourhood polygons coloured safest→highest-crime, confirming crime concentrates in a contiguous core.
2. **Crime-type composition** - each cluster's incidents split into Violent / Property / Other, revealing whether a cluster is defined by violence or by property crime.



3. **Police demand vs. crime** - per-capita scatter with an OLS trend line flags over- vs. under-served clusters relative to their crime level.



4. **Feature z-score heatmap** - shows which specific crime types define each cluster relative to the city average.

